

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
#install.packages("tidyverse")
#install.packages("lubridate")
#install.packages("zoo")
#install.packages("trend")
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric
```

```
library(trend)
library(here)
```

```
## here() starts at /home/guest/ENV 872
```

```
getwd()
```

```
## [1] "/home/guest/ENV 872"
```

```
here()
```

```
## [1] "/home/guest/ENV 872"
```

```
mytheme <- theme_bw() + theme(axis.text = element_text(color = "black"),
                             legend.position = "top",
                             legend.text = element_text(color = "black"))
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1 importing data
Ozone_2010 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"))
Ozone_2011 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"))
Ozone_2012 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"))
Ozone_2013 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"))
Ozone_2014 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"))
Ozone_2015 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"))
Ozone_2016 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"))
Ozone_2017 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"))
Ozone_2018 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"))
Ozone_2019 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"))
GaringerOzone <- bind_rows(Ozone_2010, Ozone_2011, Ozone_2012, Ozone_2013,
                           Ozone_2014,
                           Ozone_2015, Ozone_2016, Ozone_2017, Ozone_2018, Ozone_2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
#3 changing date column
GaringerOzone$Date <- mdy(GaringerOzone$Date)

#4 wrangling dataset
GaringerOzone_wrangled <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

#5 removing missing dates from dataset
Days <- as.data.frame(seq(from = as.Date("2010-01-01"),
                           to = as.Date("2019-12-31"),
                           by = "day"))
colnames(Days) <- "Date"

#6 combining datasets
GaringerOzone <- left_join(Days, GaringerOzone_wrangled)
```

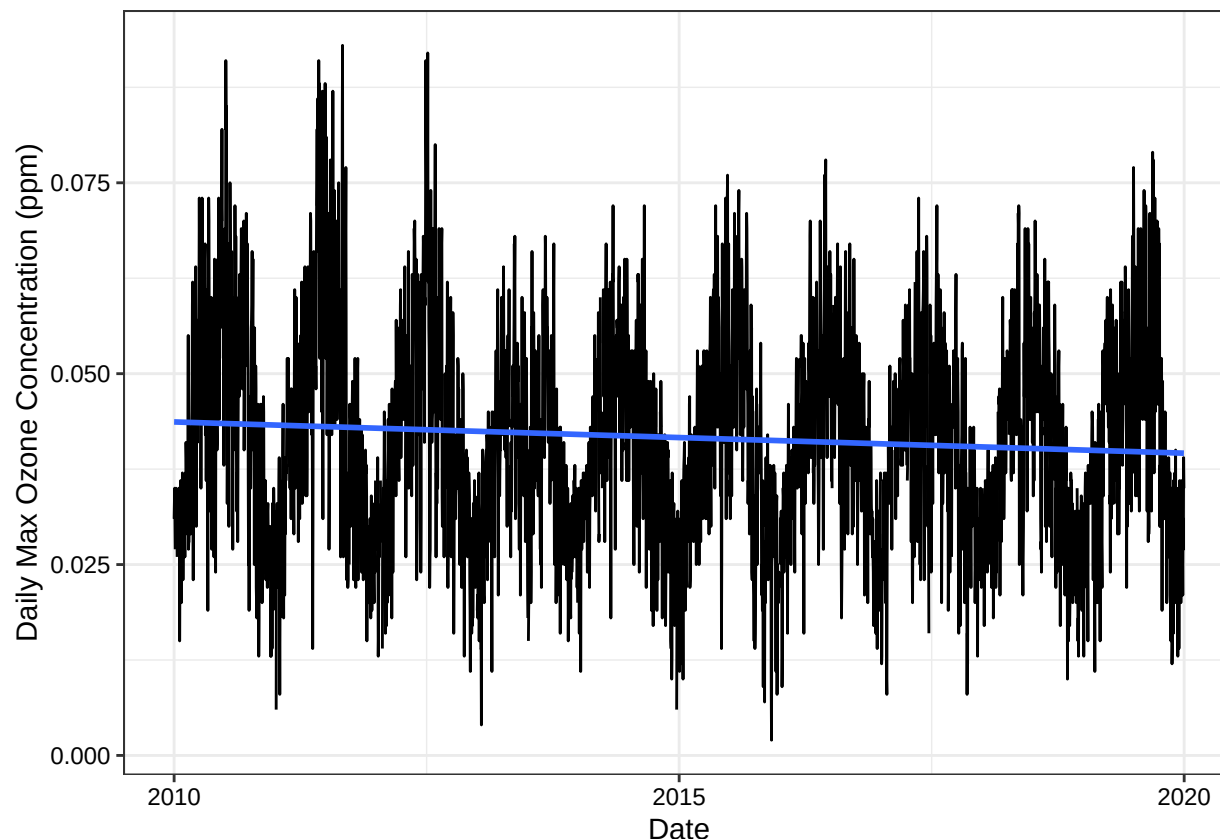
```
## Joining with 'by = join_by(Date)'
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7 plotting ozone concentrations over time
ggplot(GaringerOzone, aes(Date, Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line() + labs(y = "Daily Max Ozone Concentration (ppm)") +
  geom_smooth(method = "lm", se = F)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Answer: The plot does not suggest a strong trend in ozone concentrations over time, as the trendline shows a very weak declining slope.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8 linear interpolation to fill in missing data
GaringerOzone <- GaringerOzone %>%
  mutate(Daily.Max.8.hour.Ozone.Concentration =
    zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration))
```

Answer: We did not use a piecewise constant because that is considered the “nearest-neighbor approach”, which takes the measurement before or after the missing data and copies it. With a dataset with clear seasonal trends, this wouldn't be as accurate as applying the linear interpolation, which connects the dots instead. We did not use a spline interpolation because our data does not resemble a quadratic curve.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9 creating monthly data frame
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(Month = month(Date), Year = year(Date)) %>%
  mutate(Date2 = my(paste0(Month, "-", Year))) %>%
  group_by(Date2) %>%
  summarise(Daily.Max.8.hour.Ozone.Concentration =
    mean(Daily.Max.8.hour.Ozone.Concentration))
```

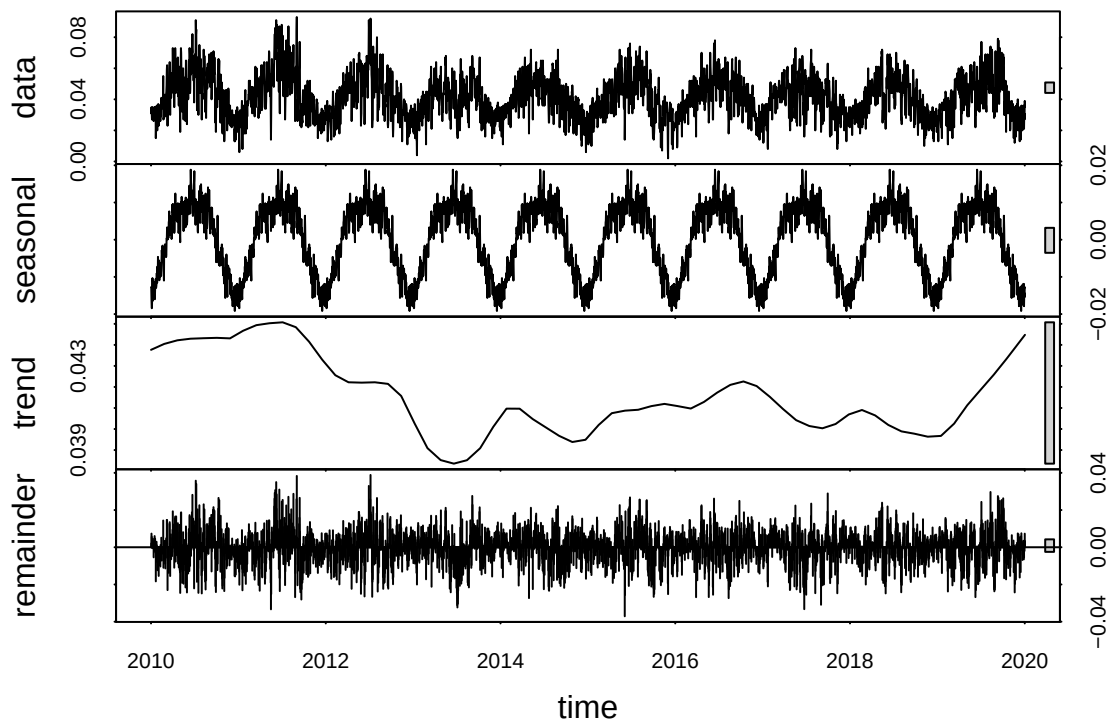
10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10 making two time series
GaringerOzone.daily.ts <- ts(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,
  start = c(2010,1),
  frequency = 365)

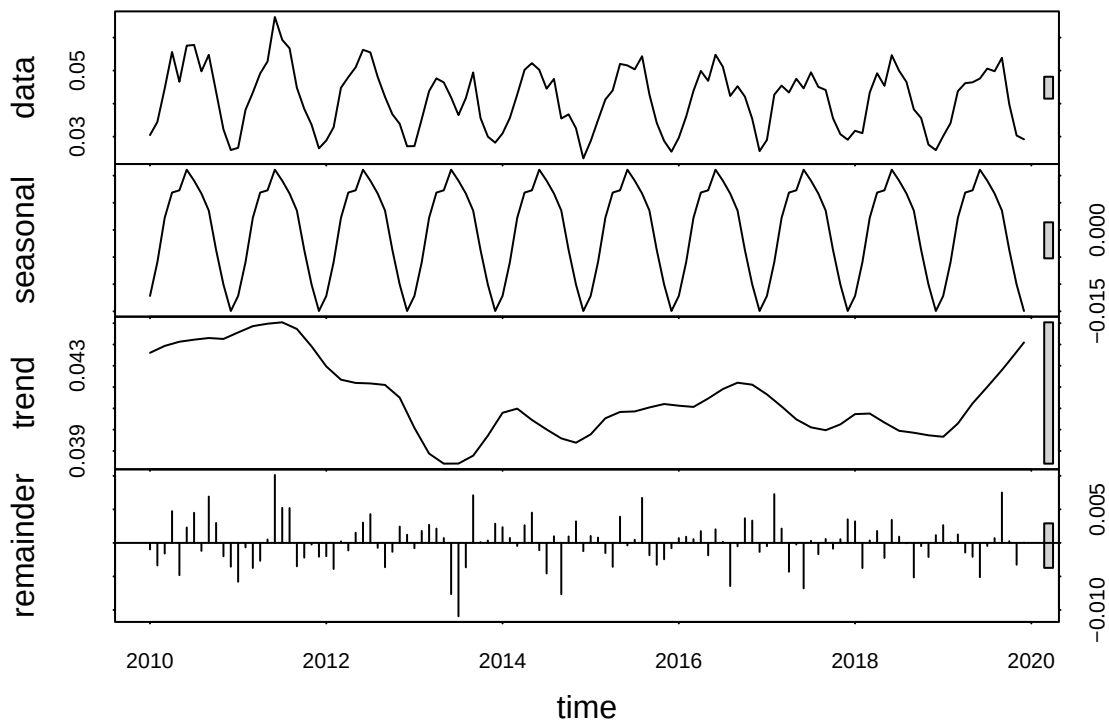
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Daily.Max.8.hour.Ozone.Concentration,
  start = c(2010,1),
  frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11 decomposing and plotting data
GaringerOzone.daily.decom <- stl(GaringerOzone.daily.ts,
  s.window = 'periodic')
plot(GaringerOzone.daily.decom)
```



```
GaringerOzone.monthly.decom <- stl(GaringerOzone.monthly.ts,
                                   s.window = 'periodic')
plot(GaringerOzone.monthly.decom)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12 monotonic trend analysis for monthly Ozone series
monthly_trend <- trend::smk.test(GaringerOzone.monthly.ts)
monthly_trend
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S varS
##    -77 1499
```

```
summary(monthly_trend)
```

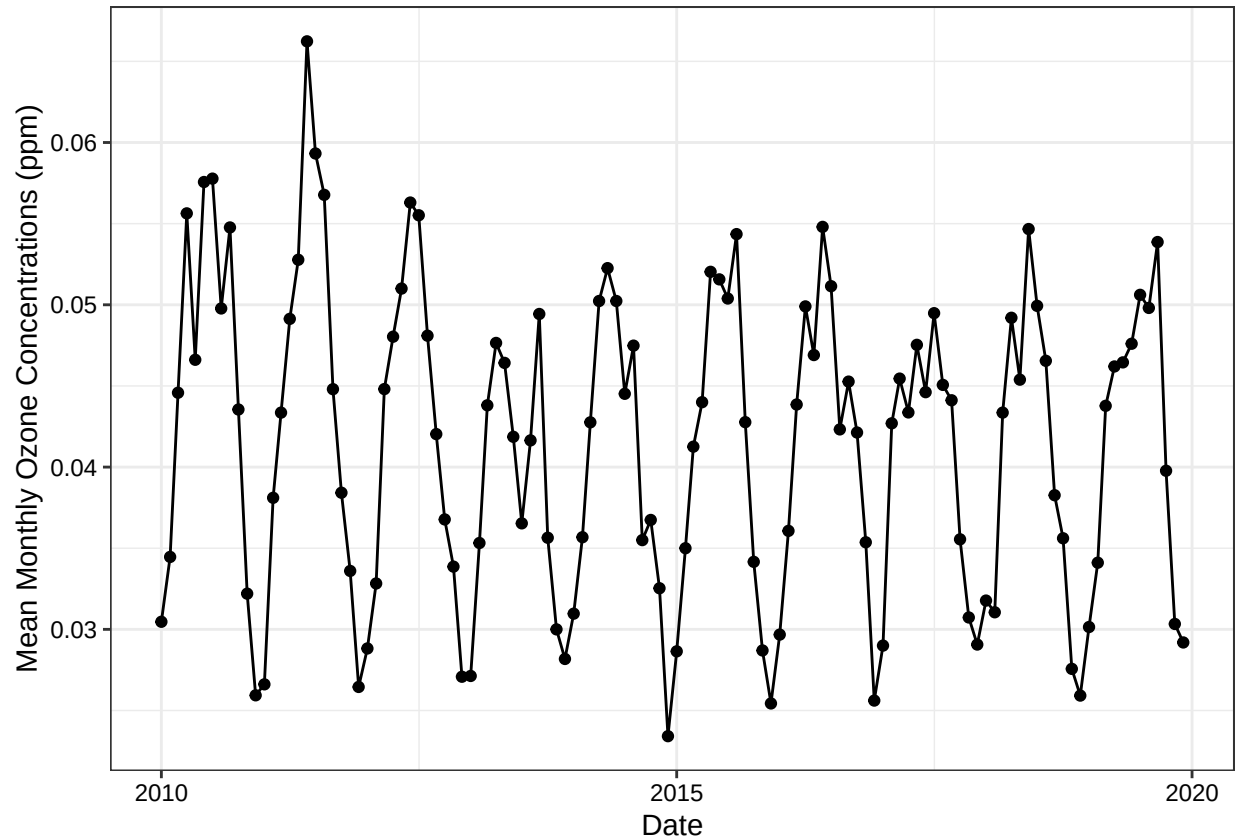
```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## alternative hypothesis: two.sided
```

```
##
## Statistics for individual seasons
##
## H0
##
##      S varS    tau      z Pr(>|z|)
## Season 1:  S = 0   15  125  0.333  1.252  0.21050
## Season 2:  S = 0   -1  125 -0.022  0.000  1.00000
## Season 3:  S = 0   -4  124 -0.090 -0.269  0.78762
## Season 4:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 5:  S = 0  -15  125 -0.333 -1.252  0.21050
## Season 6:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 7:  S = 0  -11  125 -0.244 -0.894  0.37109
## Season 8:  S = 0   -7  125 -0.156 -0.537  0.59151
## Season 9:  S = 0   -5  125 -0.111 -0.358  0.72051
## Season 10: S = 0  -13  125 -0.289 -1.073  0.28313
## Season 11: S = 0  -13  125 -0.289 -1.073  0.28313
## Season 12: S = 0   11  125  0.244  0.894  0.37109
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer: The seasonal Mann-Kendall is the most appropriate trend analysis because this test is specific for seasonal, non-parametric data, which we have in our dataset.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
#13 plotting mean monthly ozone over time
ggplot(GaringerOzone.monthly, aes(Date2, Daily.Max.8.hour.Ozone.Concentration)) +
  geom_point() + geom_line() +
  labs(x = "Date", y = "Mean Monthly Ozone Concentrations (ppm)")
```

14. To accompany your graph, summarize your results in the context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation. Question: Have ozone concentrations changed over the 2010s at this station?

Answer: Based on the time series and trend analysis of mean monthly ozone concentrations from 2010-2019, there is a very weak trend in ozone concentrations declining during this time, as the seasonal Mann-Kendall test showed an overall p-value of 0.04965. However, if the p-value is rounded it would = 0.05, and indicate no trend and acceptance of the null hypothesis. Additionally, when looking at the test statistics for individual seasons in the dataset, none of the seasons have p-values < 0.05, meaning for each individual season, the null hypothesis is accepted. Overall, the results point towards no trend of changing ozone concentrations over the 2010s.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15 subtracting seasonal component from monthly data
GaringerOzone.monthly.nonseas <- GaringerOzone.monthly.ts -
  GaringerOzone.monthly.decom$time.series[,1]
```

```
#16 Mann Kendall test on non-seasonal Ozone monthly series
```

```
monthly_noseasons <- trend::mk.test(GaringerOzone.monthly.nonseas)
monthly_noseasons
```

```
##
## Mann-Kendall trend test
##
## data:  GaringerOzone.monthly.nonseas
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## -1.179000e+03  1.943657e+05 -1.651376e-01
```

Answer: The Mann Kendall test on the non-seasonal Ozone monthly series has a p-value stronger than the one obtained with the seasonal Mann Kendall test on the complete series ($p = 0.00754$ here vs $p = 0.04965$). This suggests that when the seasonal signal is removed from the data, there is a stronger trend of declining ozone concentrations in the 2010s at this site.